

Final Project Proposal

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TigerForm: An AI Golf Swing Monitor for Comparing Player Mechanics to Tiger Woods' Swing

Project Overview

Our project is to design and develop an intelligent golf shot monitoring system that uses computer vision and supervised machine learning to analyze a golfer's swing mechanics and compare them to the swing form of Tiger Woods. The system will capture video of a user's golf swing, track key body joints, club movement, and ball trajectory, and determine how closely the swing aligns with Tiger Woods' mechanics. If the player's form differs from Tiger's, the AI will provide targeted feedback explaining what adjustments are needed to better match Tiger Woods' swing style.

The goal of this project is to create a training tool that combines sports analytics and machine learning to help golfers improve their swing by modeling one of the most technically refined swings in golf history. Rather than simply classifying a swing as "good" or "bad," the system will identify specific mechanical differences such as posture, backswing angle, hip rotation, follow-through, or club path.

Problem Statement

Golf swings are highly technical and difficult to self-correct without professional coaching or expensive launch monitor systems. Many players struggle to identify what aspects of their mechanics need improvement. By using Tiger Woods' swing as a biomechanical benchmark, this project aims to provide accessible, AI-driven coaching that offers detailed, actionable insights.

Objectives

The primary objectives of this project are:

- Build a computer vision system capable of tracking a golfer's body, club, and ball throughout a swing
- Collect and label training data from Tiger Woods' swings, including both successful and flawed shots
- Train a supervised machine learning model to recognize patterns in Tiger Woods' ideal swing form
- Compare user swing data against Tiger's swing profile
- Generate personalized corrective feedback to make the user's swing mechanics closer to Tiger's
- Prove feasibility of swing matching to an individual player

Technical Approach

1. Data Collection

We will gather a large dataset of Tiger Woods swing footage from a front camera angle and different situations (drives, irons, good shots, bad shots). Videos will be processed to extract structured movement data such as:

- Joint angles (elbows, knees, shoulders, hips)
- Spine posture and head position
- Club path and shaft angle
- Swing tempo and timing
- Ball launch direction

Pose estimation tools such as OpenPose, MediaPipe, or MoveNet will be used to convert video into numerical skeletal tracking data. Each swing instance will be labeled based on shot and form quality.

2. Supervised Learning Model

Using labeled Tiger Woods swing data, we will train supervised learning models which will classify swings as either matching or deviating from Tiger's form and identify which biomechanical variables contribute most to differences.

3. User Swing Analysis

A player will record their swing using a standard camera. The system will:

1. Extract pose and swing mechanics
2. Compare user data to the Tiger Woods benchmark model
3. Score similarity to Tiger's swing
4. Highlight deviations
5. Provide corrective recommendations

Evaluation Metrics

We will measure project success using:

- Model accuracy in identifying Tiger-like swings
- Precision and recall for good vs. poor swing classification
- Quality and usefulness of generated feedback
- User improvement in copying Tiger's form after implementing recommendations

Potential Challenges

Some expected technical challenges include:

- Variability in camera angles and lighting
- Limited access to perfectly labeled Tiger Woods swing data
- Translating raw biomechanical differences into useful coaching advice

Expected Outcome

By the end of this project, we expect to produce a functional prototype that can analyze a golfer's swing, compare it to Tiger Woods' biomechanics, and provide personalized AI coaching recommendations. This system could serve as a way for fans to model their golf game after their favorite players while demonstrating how machine learning and computer vision can be applied in sports performance training.